PROJECT GUIDELINE

Subject: DATA SCIENCE & ANALYTICS

Groups & Theme Allocation:

https://docs.google.com/spreadsheets/d/1PKF-tG-xgkohzPTukdya1TzvqVCrDt-qETTMCiJIJR4/edit?usp=sharing

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IDEAS FOR YOUR THEMES:

Theme A

Banking Analytics

Loan default prediction

ATM transaction analysis

Digital banking adoption

Branch profitability

Customer churn

Credit card usage

Theme B

Investment Analytics

Stock trend analysis

Portfolio optimization

Sector comparison

Mutual fund performance

Risk-return analysis

Crypto volatility

Gold vs Equity

Theme C

Insurance Analytics

Claim prediction

Fraud detection

Health insurance analytics

Motor insurance

Customer segmentation

Premium optimization

Theme D

FinTech Analytics

UPI adoption

Wallet usage

BNPL analysis

Digital lending

Payment trends

FinTech customer behaviour

Theme E

Supply Chain Analytics

Demand forecasting

Inventory optimization

Supply chain analytics

Price Sensitivity/Elasticity of Supply Chain

Theme F

Marketing Analytics

Sales forecasting

Marketing analytics

Advertising Analytics

Pricing analytics

Customer lifetime value

Theme G

Public Policy Analytics

GDP

Inflation

Unemployment

Electricity

Agriculture

Transport

Climate

Urbanization

Financial inclusion

Theme H

Sports Analytics

Football - FIFA

Cricket - ICC World Cup

Cricket - IPL

Basketball - NBA

Tennis - Grand Slam

F1 Racing

Athletics

Olympics

Commonwealth Games

Performance/ Strategy Analytics

Theme I

Educational Analytics

Paper Leak Problem Analysis

Literacy Rate - Unemployment Correlation

Student Dropout Prediction Models

Impact of Online vs. Offline Learning

Predicting Placement Success

Scholarship Allocation Efficiency

Teacher Evaluation Analytics

School Infrastructure and Academic Performance

Gender Gap in STEM Education

Effectiveness of Remedial Coaching

Theme J

Healthcare Analytics

Health Insurance in India

Trends in Healthcare

Patient Readmission Prediction

Hospital Resource & Bed Capacity Management

Disease Outbreak Pattern Analysis

Telemedicine Adoption and Effectiveness

Healthcare Cost Optimization

Chronic Disease Management Trends

Wait Time Analysis in Outpatient Departments

Medical Staff Allocation and Efficiency

Examples of project titles:

Banking

Predicting Loan Default Risk Using Customer Financial Data

Stock Market

Forecasting Reliance Share Price using Time Series

Investment

Risk Return Comparison of Nifty 50 Sectors

FinTech

UPI Transaction Growth Analysis across Indian States

Healthcare

Hospital Admission Forecasting

Education

Student Performance Analytics

Sports

IPL Performance Analytics

STAGE-WISE PROJECT WORK:

Stage 1

Preparing a Problem Statement

Examples:

- Why are loan defaults increasing?
- Which customers should receive premium credit cards?
- Which mutual fund performs consistently?
- Which UPI app is preferred?
- Which districts have the highest insurance penetration?
- Can electricity consumption predict industrial production?
- Does rainfall affect crop insurance claims?
- Why are bank frauds increasing?

Note: The problem should be business-oriented. Not technical.

Stage 2

Research

Study:

- Industry
- Company
- Existing situation
- Background

Summarise in 2-3 pages.

Mention all your sources for research

Stage 3

Data Collection

1. Conduct Surveys in your vicinity to collect primary data. Maintain Google forms/Excel Sheets.

2. You are free to use any sources as per your requirement.

- Kaggle
- RBI
- NSE
- BSE
- Yahoo Finance
- World Bank
- Government Open Data
- Company Annual Reports
- Surveys
- Web scraping (optional)

(Not every project would require huge datasets)

3. Some may combine:

Survey + Public data

Stage 4

Data Cleaning

You must show:

Missing values

Outliers

Duplicates

Formatting

Normalization

This should become one chapter.

Stage 5

Exploratory Data Analysis

Mandatory.

Every project must contain

- Histograms
- Bar charts
- Line graphs
- Scatter plots
- Correlation heatmap
- Pivot tables
- Trend analysis

Stage 6

Apply Analytics

Depending on projectL

Regression

OR

Time Series

OR

Clustering

OR

Forecasting

OR

Monte Carlo

OR

Risk Analysis

OR

Hypothesis Testing

OR

Customer Segmentation

Note: Every project need not use every technique. Choose suitable tools.

Stage 7

Insights

Not observations. Insights.

Example:

Observation:

Loan defaults increased 15%.

Insight:

Customers below ₹4 lakh income contribute 70% of defaults despite representing only 40% of borrowers.

Insights should be valuable.

Stage 8

Recommendations

Business recommendations. Suggest practical solutions.

You might recommend:

Pricing

Marketing

Risk controls

Investment

Operations

Customer segmentation

Automation

Stage 9

Executive Summary

One page. Should be like consulting reports.

Suggested Report Structure (20–30 pages):

1. Cover Page
2. Certificate
3. Acknowledgement
4. Executive Summary
5. Problem Statement
6. Objectives
7. Industry Background
8. Literature Review
9. Dataset Description
10. Methodology
11. Data Cleaning
12. Exploratory Data Analysis
13. Statistical Analysis
14. Results
15. Business Insights
16. Recommendations
17. Limitations
18. Future Scope
19. References
20. Appendix

Tentative Evaluation Rubric:

Component	Marks
Problem Statement	10
Data Collection	10
Data Cleaning	10
Analytical Techniques	20
Insights & Interpretation	15
Business Recommendations	10
Report Quality	10
Presentation & Viva	15